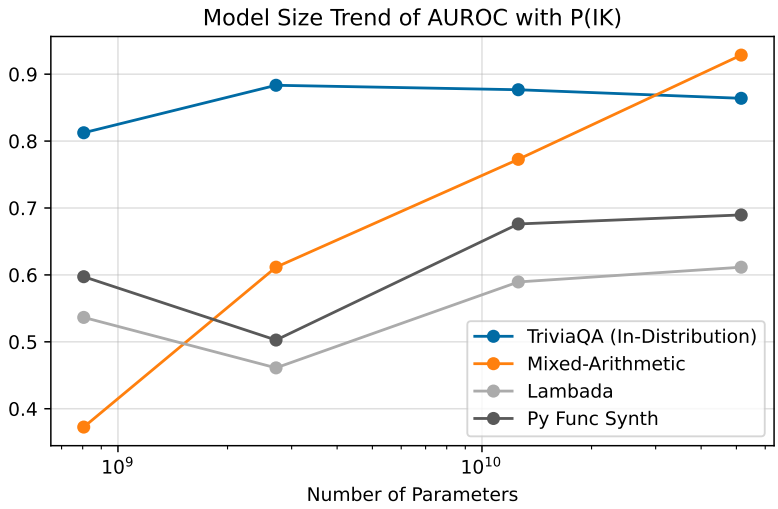

Figure 14 Model size trend of AUROC with P(IK), when training on only TriviaQA. We generally observe increasing AUROC as model size increases for all three out-of-distribution evals, suggesting that larger P(IK) classifiers are better at generalization.

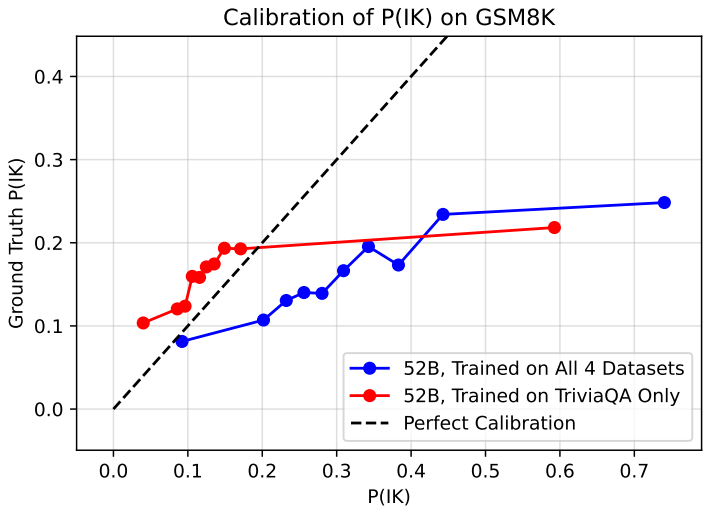

Figure 15 Calibration on GSM8k when training on only TriviaQA vs training on TriviaQA, LAMBADA, Mixed-Arithmetic, and Python Function Synthesis. We see that calibration does improve when training on more diverse data, rather instead of being mostly underconfident the model becomes overconfident. Note that GSM8k was significantly out of the training distribution in all cases.

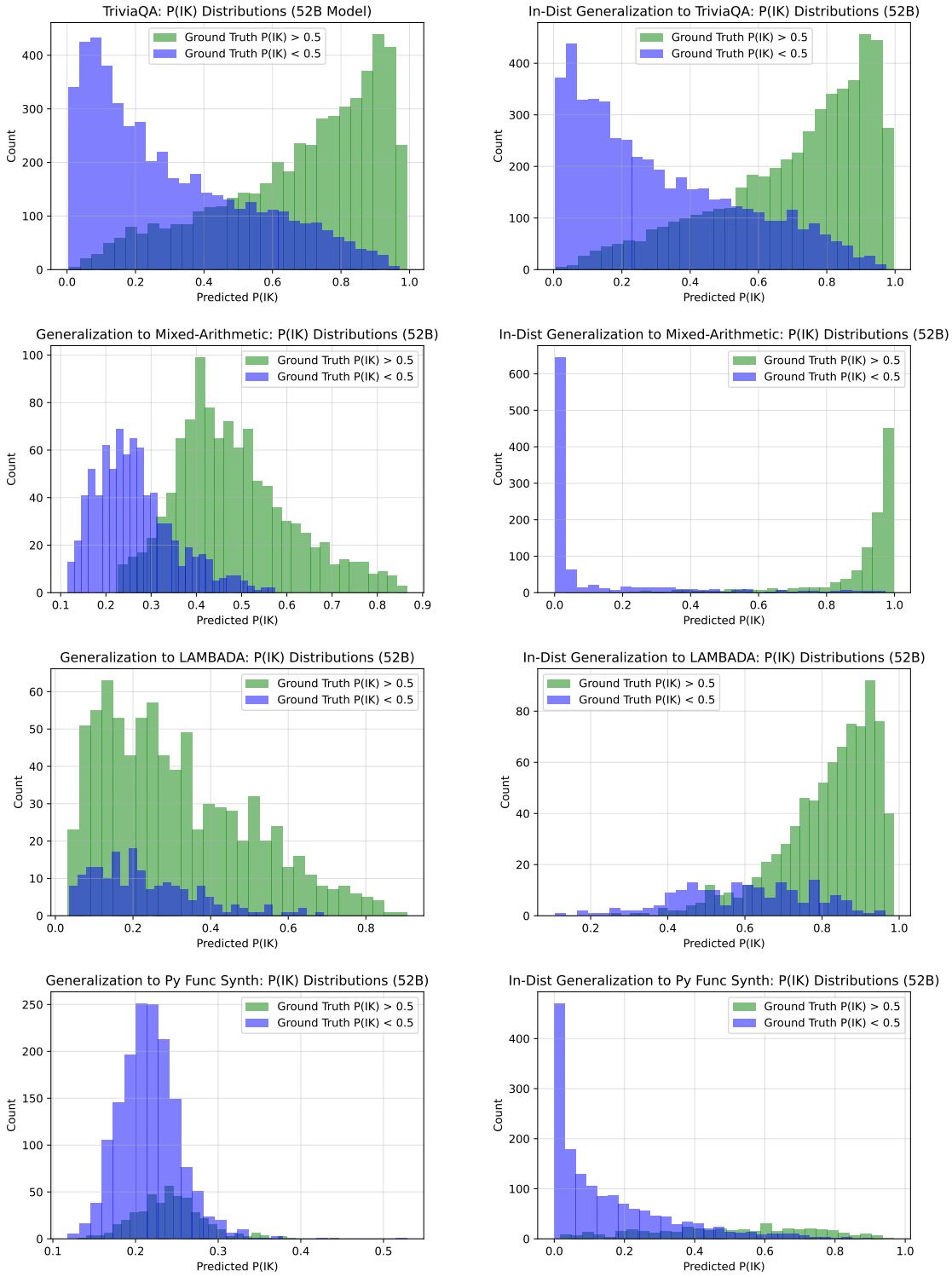

Figure 16 Generalization of P(IK). We trained P(IK) classifiers on just TriviaQA and on TriviaQA, Arithmetic, Python Function Synthesis, and LAMBADA. The left side of this figure includes distributions of P(IK) from a 52B classifier that was trained on just TriviaQA, while the right side includes distributions of P(IK) from a 52B classifier that was trained on all of these data distributions. We observe nontrivial generalization from TriviaQA to the other tasks, but training on the other tasks improves performance greatly.

	Training on TriviaQA (AUROC / Brier Score)	Training on All Tasks Except GSM8K (AUROC / Brier Score)
TriviaQA	0.864 / 0.151	0.873 / 0.145
Mixed-Arithmetic	0.928 / 0.194	0.987 / 0.042
LAMBADA	0.606 / 0.431	0.853 / 0.108
Python Func Synth.	0.687 / 0.164	0.881 / 0.109
GSM8K [6]	0.624 / 0.200	0.752 / 0.121

Table 1 AUROC and Brier Scores of P(IK) on a variety of tasks, comparing training on just TriviaQA and training on everything. All values are computed using 52B parameter classifiers. All AUROC scores here are computed by comparing the model's predicted P(IK) against the binarized label: 'IK' if the ground truth P(IK) > 0.5, and 'IDK' otherwise. Even when we only train on TriviaQA, we see decent generalization to other tasks. However, training on everything does help across all tasks.

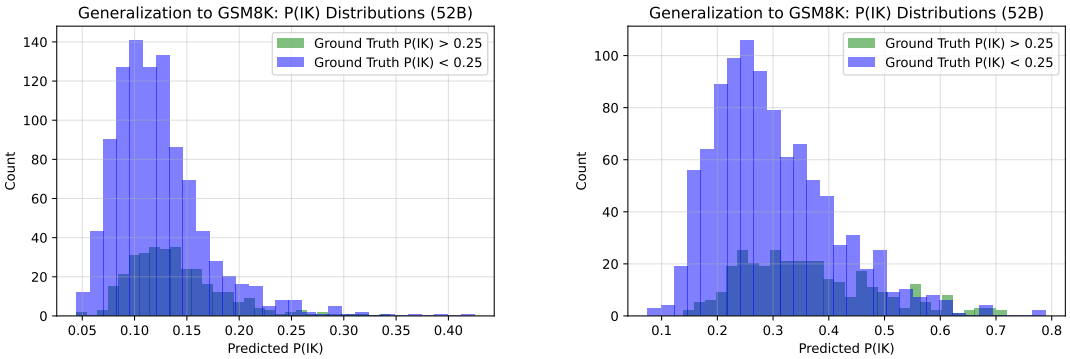

Figure 17 Left: P(IK) scores on GSM8k when training on only TriviaQA. Right: P(IK) scores on GSM8k when training on TriviaQA, LAMBADA, Mixed-Arithmetic, and Python Function Synthesis. Note the threshold for Ground Truth P(IK) is 0.25 here, in contrast to 0.5 in the other histograms from Figure 16 and Table 1

5.3 P(IK) Generalizes to Account for Source Materials

If we consider a fairly obscure question like

```
What state's rodeo hall of fame was established in 2013?
```

then P(IK) appropriately predicts a low value, specifically 18% for a 52B model. However, if we prepend a Wikipedia article on the Idaho Rodeo Hall of Fame to the context:

```
Wikipedia: The Idaho Rodeo Hall of Fame was established as a 501 (c)(3) non-
    profit organization on May 6, 2013. Lonnie and Charmy LeaVell are the
    founders of the organization. The actual charitable nonprofit status was
    received from the IRS on February 19, 2014. The IRHF hosts a reunion and
    induction ceremony annually every October. The Idaho Hall of Fame preserves
    and promotes the Western lifestyle and its heritage. The hall exists to
    dedicate the men and women in rodeo who contribute to ranching and farming
    through their sport. It also extends its reach to continue these western
    ways to the youth in the communities to ensure that these traditions
    continue for many generations. In 2015, the hall was awarded the Historic
    Preservation Recognition Award by National Society of the Daughters of the
    American Revolution.

What state's rodeo hall of fame was established in 2013?
```